

Automated Analyst Report

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Query: Analyse supplier performance across all product types. Identify which product categories have the highest defect rates, which shipping carriers offer the best revenue-to-cost ratio, flag any SKUs with failed inspection results, and highlight stockout risk where lead times are long relative to stock levels. report findings must be backed by solid proof from within the data

****EXECUTIVE SUMMARY**** Haircare products exhibit the highest average defect rate (2.48 %) while Carrier C delivers the strongest revenue-to-shipping-cost performance (\$1,138.56 revenue per shipping dollar). Thirty-six SKUs failed inspection, with several (e.g., SKU 42, SKU 65) showing defect rates above 4.9 % and low on-hand stock. Stock-out exposure is most acute for SKU 34 (risk score 13) and SKU 68 (risk score 8), indicating lead-time pressures that outpace current inventory.

****KEY FINDINGS**** - ****Product type defect hierarchy:**** Haircare = 2.483 % > Skincare = 2.335 % > Cosmetics = 1.919 % (average defect rates). - ****Carrier efficiency:**** Carrier C generates the highest revenue per shipping dollar (\$1,138.56), followed by Carrier B (\$1,055.70) and Carrier A (\$917). Carrier C also maintains a moderate avg. defect rate (2.6 %) compared with Carrier A (2.74 %). - ****Failed inspection SKUs:**** All 36 listed SKUs failed inspection; the top five by defect severity are SKU 42 (4.94 %), SKU 65 (4.91 %), SKU 50 (4.75 %), SKU 3 (4.75 %), and SKU 73 (4.62 %). SKU 68 has a defect rate of 1.98 % but zero stock on hand. - ****Revenue contribution:**** Skincare leads revenue (\$241,628), followed by Haircare (\$174,455) and Cosmetics (\$161,521). Gross-margin ratios are uniformly high (> 99 %) across all categories. - ****Stock-out risk scores:**** SKU 34 (score 13), SKU 68 (8), SKU 24 (5.2), SKU 2 (5.0) and SKU 33 (3.4) rank highest, reflecting lead times that far exceed available inventory (e.g., SKU 34 has 1 unit in stock with a 26-day lead time).

****RISKS AND RECOMMENDATIONS**** The convergence of elevated defect rates in haircare, a concentration of failed inspection SKUs with low inventory, and high stock-out risk scores threatens service levels and margin protection. Prioritize corrective action on the 36 failed SKUs—especially those with defect rates > 4 % and stock ≤ 20 units—and re-evaluate carrier contracts to lock in Carrier C's superior cost efficiency while monitoring its defect performance. Simultaneously, implement safety-stock buffers for the top-risk SKUs (e.g., SKU 34, SKU 68) and accelerate lead-time reduction initiatives (alternative routing or local sourcing) to mitigate imminent stock-out exposure.